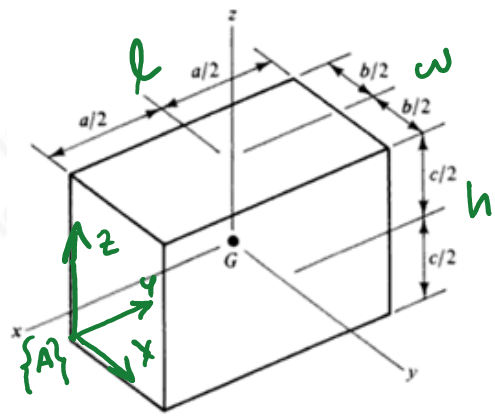
Problem 1: For the rectangular body shown, let the moment of inertia around be known known around C.G. where frame $\{G\}$ is locted. Use parallel axis therem to evaluate the inertia tensor with resepct to the a frame located at the lower left corner of it. Consider uniform density.

FROM SLIDE 7, THE INERTIA TENSOR WITH RESPECT TO THE CENTER OF MASS IS:



$V = abc$

$I_{xx} = \frac{1}{12}m(b^2+c^2)$

$I_{yy} = \frac{1}{12}m(a^2+c^2)$

$I_{zz} = \frac{1}{12}m(a^2+b^2)$

$$^{G}I = \frac{m}{12}\begin{bmatrix} \ell^2+h^2 & 0 & 0 \\ 0 & w^2+h^2 & 0 \\ 0 & 0 & w^2+\ell^2 \end{bmatrix}$$

THE TRANSLATION OF THE INERTIA TENSOR FROM FRAME $\{G\}$ TO FRAME $\{A\}$ IS GIVEN BY

$X_A = P_{x_{G\to A}} = -\frac{w}{2}$

$Y_A = P_{y_{G\to A}} = -\frac{\ell}{2}$

$Z_A = P_{z_{G\to A}} = -\frac{h}{2}$

FROM SLIDE 12,

$$^{A}I_{xx} = {}^{G}I_{xx} + m(Y_A^2 + Z_A^2) = \frac{1}{12}m(\ell^2+h^2) + m\left(\left(-\frac{\ell}{2}\right)^2 + \left(-\frac{h}{2}\right)^2\right) = \frac{1}{3}m(\ell^2+h^2)$$

SIMILARLY, $^{A}I_{yy} = \frac{1}{3}m(w^2+h^2)$ $\quad$ $^{A}I_{zz} = \frac{1}{3}m(\ell^2+w^2)$

PRODUCTS OF INERTIA:

$$^{A}I_{xy} = {}^{G}I_{xy} + m\,X_A Y_A = 0 + m\left(\frac{-w}{2}\right)\left(-\frac{\ell}{2}\right) = \frac{m}{4}w\ell$$

SIMILARLY, $^{A}I_{xz} = \frac{m}{4}wh$ $\quad$ $^{A}I_{yz} = \frac{m}{4}\ell h$

THE INERTIA TENSOR IS:

$$^{A}I = \begin{bmatrix} \frac{m}{3}(\ell^2+h^2) & -\frac{m}{4}w\ell & -\frac{m}{4}wh \\ -\frac{m}{4}w\ell & \frac{m}{3}(w^2+h^2) & -\frac{m}{4}\ell h \\ -\frac{m}{4}wh & -\frac{m}{4}\ell h & \frac{m}{3}(\ell^2+w^2) \end{bmatrix}$$

NOTE: ^{A}I IS ALWAYS A POSITIVE DEFINITE MATRIX AND IS SYMMETRIC. ALL EIGENVALUES OF ^{A}I ARE POSITIVE.

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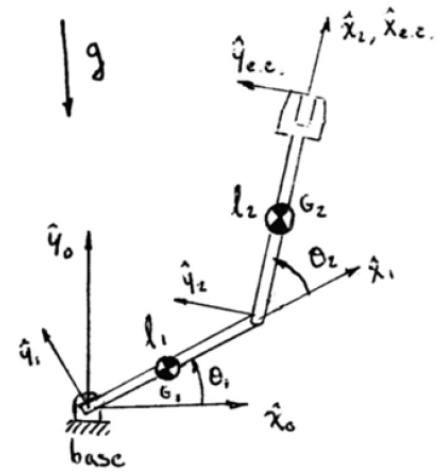
Problem 2: Compute the closed form dynamic equation of the two revolute jointed planar manipulator using the Lagrangian formulation.



Characteristics of the system:

- Assume the links to be slender bars.
- Let the center of mass be in the middle of each link.
- Assume the base is not rotating.
- No forces or moments are being applied at the end-effector.

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \tau_i$$

$$L = k_1 + u_1 + k_2 - u_2$$

$$k_1 = \frac{1}{2}\,{}^1v_{G1}{}^T m_1\,{}^1v_{G1} + \frac{1}{2}\,{}^1\omega_1{}^T\,{}^0I_1\,{}^1\omega_1$$

$${}^{G_1}I_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

$${}^1\omega_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$${}^1v_{G1} = {}^1\omega_1 \times P_{G1} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{bmatrix} l_1/2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ l_1\dot{\theta}_1/2 \\ 0 \end{bmatrix}$$

$$\frac{1}{2}\,{}^0v_{G1}{}^T m_1\,{}^0v_{G1} = \frac{1}{2}m_1\begin{bmatrix} 0 & l_1\dot{\theta}_1/2 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ l_1\dot{\theta}_1/2 \\ 0 \end{bmatrix} = \frac{m_1(l_1\dot{\theta}_1)^2}{8}$$

$${}^1\omega_1{}^T\,{}^{G_1}I_1\,{}^1\omega_1 = \begin{bmatrix} 0 & 0 & \dot{\theta}_1 \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} = I_{zz}\dot{\theta}_1^2$$

$$k_1 = \frac{m_1(l_1\dot{\theta}_1)^2}{8} + \frac{I_{zz}\dot{\theta}_1^2}{2}$$

$$u_1 = -m_1 g^T \, {}^0P_{G1}$$

$$= -m_1 \begin{bmatrix} 0 & -g & 0 \end{bmatrix} \begin{bmatrix} \frac{l_1 c_1}{2} \\ \frac{l_1 s_1}{2} \\ \\ \end{bmatrix} = m_1 g \frac{l_1 s_1}{2}$$

$$k_2 = \frac{1}{2} \, {}^2v_{G2}{}^T m_2 \, {}^2v_{G2} + \frac{1}{2} \, {}^2\omega_2{}^T \, {}^{G_2}I_2 \, {}^2\omega_2$$

$${}^{G_2}I_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

$${}^2\omega_2 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix}$$

$${}^2v_{G2} = {}^2R_1 \, {}^1v_1 + {}^2\omega_2 \times P_{G2} = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{bmatrix} l_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \times \begin{bmatrix} l_2/2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} s_2 l_1 \dot{\theta}_1 \\ c_2 l_1 \dot{\theta}_1 + l_2(\dot{\theta}_1 + \dot{\theta}_2)/2 \\ 0 \end{bmatrix}$$

$$\frac{1}{2} \, {}^2v_{G2}{}^T m_2 \, {}^2v_{G2} = \frac{1}{2} m_2 \begin{bmatrix} s_2 l_1 \dot{\theta}_1 & c_2 l_1 \dot{\theta}_1 + l_2(\dot{\theta}_1 + \dot{\theta}_2)/2 & 0 \end{bmatrix} \begin{bmatrix} s_2 l_1 \dot{\theta}_1 \\ c_2 l_1 \dot{\theta}_1 + l_2(\dot{\theta}_1 + \dot{\theta}_2)/2 \\ 0 \end{bmatrix}$$

$$= \frac{1}{2} m_2 [(s_2 l_1 \dot{\theta}_1)^2 + (c_2 l_1 \dot{\theta}_1 + l_2(\dot{\theta}_1 + \dot{\theta}_2)/2)^2] = m_2((l_1 \dot{\theta}_1)^2/2 + c_2 l_1 \dot{\theta}_1 l_2(\dot{\theta}_1 + \dot{\theta}_2)/2 + (l_2(\dot{\theta}_1 + \dot{\theta}_2))^2/8)$$

$$\frac{1}{2} \, {}^2\omega_2{}^T \, {}^{G2}I_2 \, {}^2\omega_2 = \begin{bmatrix} 0 & 0 & \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy2} & 0 \\ 0 & 0 & I_{zz2} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} = \frac{1}{2} I_{zz2} (\dot{\theta}_1 + \dot{\theta}_2)^2$$

$$k_2 = m_2(l_1 \dot{\theta}_1)^2/2 + m_2 c_2 l_1 l_2 \dot{\theta}_1^2/2 + m_2 c_2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2/2 + \frac{m_2(l_2 \dot{\theta}_1)^2}{8} + \frac{m_2(l_2 \dot{\theta}_2)^2}{8} + \frac{m_2 l_2 \dot{\theta}_1 \dot{\theta}_2}{4} + \frac{I_{zz2} \dot{\theta}_2^2}{2} + \frac{I_{zz2} \dot{\theta}_1^2}{2} + I_{zz2} \dot{\theta}_1 \dot{\theta}_2$$

$$u_2 = -m_2 g^T \, {}^0P_{G2}$$

$$= -m_2 \begin{bmatrix} 0 & -g & 0 \end{bmatrix} \begin{bmatrix} l_1 c_1 + \frac{l_2 c_{12}}{2} \\ l_1 s_1 + \frac{l_2 s_{12}}{2} \\ 0 \end{bmatrix} = m_2 g (l_1 c_1 + \frac{l_2 c_{12}}{2})$$

$$L = k_1 + u_1 + k_2 - u_2 = \frac{m_1 (l_1 \dot{\theta}_1)^2}{8} + \frac{I_{zz1} \dot{\theta}_1^2}{2} + m_2 (l_1 \dot{\theta}_1)^2 / 2 + m_2 c_2 l_1 l_2 \dot{\theta}_1^2 + m_2 c_2 l_1 \dot{\theta}_1 l_2 \dot{\theta}_2 / 2 + \frac{m_2 (l_2 \dot{\theta}_1)^2}{8}$$
$$+ \frac{m_2 (l_2 \dot{\theta}_2)^2}{8} + \frac{m_2 l_2 \dot{\theta}_1 \dot{\theta}_2}{4} + \frac{I_{zz2} \dot{\theta}_1^2}{2} + \frac{I_{zz2} \dot{\theta}_2^2}{2} + I_{zz2} \dot{\theta}_1 \dot{\theta}_2 - m_1 g (\frac{l_1 s_1}{2}) - m_2 g (l_1 s_1 + \frac{l_2 s_{12}}{2})$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \tau_i$$

$for \, \theta_1$

$$\frac{\partial L}{\partial \dot{\theta}_1} = \frac{m_1 l_1^2 \dot{\theta}_1}{4} + I_{zz1} \dot{\theta}_1 + m_2 l_1^2 \dot{\theta}_1 / 2 + m_2 c_2 l_1 l_2 \dot{\theta}_1 + m_2 c_2 l_1 l_2 \dot{\theta}_2 / 2 + \frac{m_2 l_2^2 \dot{\theta}_1}{4} + \frac{m_2 l_2^2 \dot{\theta}_2}{4} + I_{zz2} \dot{\theta}_1 + I_{zz2} \dot{\theta}_2$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} = \frac{m_1 l_1^2 \ddot{\theta}_1}{4} + I_{zz1} \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 / 2 - m_2 s_2 \dot{\theta}_2 l_1 l_2 \dot{\theta}_1 + m_2 c_2 l_1 l_2 \ddot{\theta}_1 - m_2 s_2 l_1 l_2 \dot{\theta}_2^2 / 2 + m_2 c_2 l_1 l_2 \ddot{\theta}_2 / 2 + \frac{m_2 l_2^2 \ddot{\theta}_1}{4} + \frac{m_2 l_2^2 \ddot{\theta}_2}{4}$$
$$+ I_{zz2} \ddot{\theta}_1 + I_{zz} \ddot{\theta}_2$$

$$-\frac{\partial L}{\partial \theta_1} = m_1 g (\frac{l_1 c_1}{2}) + m_2 g (l_1 c_1 + \frac{l_2 c_{12}}{2})$$

$$\tau_1 = \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_1} - \frac{\partial L}{\partial \theta_1} = \frac{m_1 l_1^2 \ddot{\theta}_1}{4} + I_{zz1} \ddot{\theta}_1 + m_2 l_1^2 \ddot{\theta}_1 / 2 + m_2 c_2 l_1 l_2 \ddot{\theta}_1 + m_2 c_2 l_1 l_2 \ddot{\theta}_2 / 2 + \frac{m_2 l_2^2 \ddot{\theta}_1}{4} + \frac{m_2 l_2^2 \ddot{\theta}_2}{4} + I_{zz2} \ddot{\theta}_1$$
$$+ I_{zz} \ddot{\theta}_2 + m_1 g (\frac{l_1 c_1}{2}) + m_2 g (l_1 c_1 + \frac{l_2 c_{12}}{2})$$

$for\ \theta_2$

$$\frac{\partial L}{\partial \dot{\theta}_2} = m_2 c_2 l_1 l_2 \dot{\theta}_1 / 2 + \frac{m_2 l_2^2 \dot{\theta}_2}{4} + \frac{m_2 l_2 \dot{\theta}_1}{4} + I_{zz2}\dot{\theta}_2 + I_{zz2}\dot{\theta}_1$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_2} = -m_2 s_2 \dot{\theta}_2 l_1 l_2 \dot{\theta}_1 / 2 + m_2 c_2 l_1 l_2 \ddot{\theta}_1 / 2 + \frac{m_2 l_2^2 \ddot{\theta}_2}{4} + \frac{m_2 l_2 \ddot{\theta}_1}{4} + I_{zz2}\ddot{\theta}_2 + I_{zz2}\ddot{\theta}_1$$

$$-\frac{\partial L}{\partial \theta_2} = +m_2 s_2 l_1 l_2 \dot{\theta}_1^2 + m_2 s_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 / 2 + m_2 g \frac{l_2 c_{12}}{2}$$

$$\tau_2 = \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}_2} - \frac{\partial L}{\partial \theta_2} = m_2 c_2 l_1 l_2 \ddot{\theta}_1 / 2 + \frac{m_2 l_2^2 \ddot{\theta}_2}{4} + \frac{m_2 l_2 \ddot{\theta}_1}{4} + I_{zz2}\ddot{\theta}_2 + I_{zz2}\ddot{\theta}_1 + m_2 g \frac{l_2 c_{12}}{2}$$

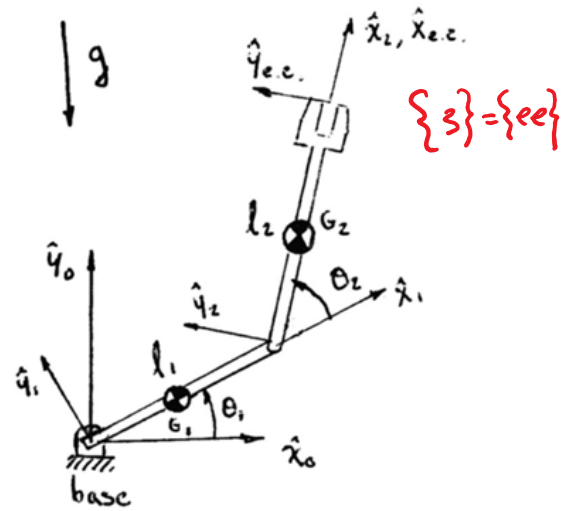
$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} \frac{m_1 l_1^{\ 2}}{4} + I_{zz1} + m_2 l_1 / 2 + m_2 c_2 l_1 l_2 + \frac{m_2 l_2^{\ 2}}{4} + I_{zz2} & m_2 c_2 l_1 l_2 / 2 + \frac{m_2 l_2^{\ 2}}{4} + I_{zz2} \\ m_2 c_2 l_1 l_2 / 2 + \frac{m_2 l_2}{4} + I_{zz2} & \frac{m_2 l_2^2}{4} + I_{zz2} \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix}$$

$$+ \begin{bmatrix} -m_2 s_2 \dot{\theta}_2 l_1 l_2 \dot{\theta}_1 - m_2 s_2 l_1 l_2 \dot{\theta}_2^2 / 2 \\ m_2 s_2 l_1 l_2 \dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} m_1 (\frac{l_1 c_1}{2}) + m_2 (l_1 c_1 + \frac{l_2 c_{12}}{2}) \\ m_2 \frac{l_2 c_{12}}{2} \end{bmatrix} g$$

Problem 3: Compute the closed form dynamic equation of the two revolute jointed planar manipulator using the Newton-Euler recursive formulation.

Characteristics of the system:

- Assume the links to be slender bars
- Let the center of mass be in the middle of each link
- Assume the base is not rotating
- No forces or moments are being applied at the end-effector



THE HOMOGENEOUS MATRICES OF THIS MANIPULATOR ARE SHOWN BELLOW, AS WELL AS THE ROTATION MATRICES AND POSITION VECTORS.

$$ {}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^0\vec{P}_1 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}, \quad {}^1\vec{P}_{G_1} = \begin{Bmatrix} l_1/2 \\ 0 \\ 0 \end{Bmatrix}, \quad {}^0_1R = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^1_0R = \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

$$ {}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1\vec{P}_2 = \begin{Bmatrix} l_1 \\ 0 \\ 0 \end{Bmatrix}, \quad {}^2\vec{P}_{G_2} = \begin{Bmatrix} \frac{l_2}{2} \\ 0 \\ 0 \end{Bmatrix}, \quad {}^0_1R = \begin{bmatrix} c_2 & -s_2 & 0 \\ s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^1_0R = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

$$ {}^2_3T = \begin{bmatrix} 1 & 0 & 0 & l_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^2_3R = {}^3_2R = I_{3\times3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad {}^2\vec{P}_3 = \begin{Bmatrix} l_2 \\ 0 \\ 0 \end{Bmatrix} $$

FROM SLIDE I, THE INERTIA TENSOR OF THE LINKS (WHICH ARE SLENDER BARS) ARE:

$$ {}^{G_1}I_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy_1} & 0 \\ 0 & 0 & I_{zz_1} \end{bmatrix} = \frac{1}{12} m_1 l_1^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad {}^{G_2}I_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy_2} & 0 \\ 0 & 0 & I_{zz_2} \end{bmatrix} = \frac{1}{12} m_2 l_2^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

NOTE: $I_{xx_1} = I_{xx_2} = 0$

✓ GIVEN THAT THE BASE IS NOT ROTATING, ${}^0\vec{\omega}_0 = 0 = {}^0\dot{\vec{\omega}}_0$

✓ APPLYING THE GRAVITY EFFECT, ${}^0\dot{\vec{v}}_0 = \begin{Bmatrix} 0 \\ g \\ 0 \end{Bmatrix}$ (IN THE FIGURE, $\hat{Y}_0$ IS THE VERTICAL AXIS)

$$ {}^3\vec{f}_3 = {}^3\vec{n}_3 = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix} $$

OUTWARD KINEMATICS $\quad i = 0 \rightarrow 1$

USING SLIDE #19

$i = 0$

$$ {}^{1}\vec{\omega}_1 = {}^{1}_{0}R\,{}^{0}\vec{\omega}_0 + \dot{\theta}_1\,{}^{1}\hat{z}_1 = \vec{0} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} $$

$$ {}^{1}\dot{\vec{\omega}}_1 = {}^{1}_{0}R\,{}^{0}\dot{\vec{\omega}}_0 + {}^{1}_{0}R\,{}^{0}\vec{\omega}_0 \times \dot{\theta}_1\,{}^{1}\hat{z}_1 + \ddot{\theta}_1\,{}^{1}\hat{z}_1 = \vec{0} + \vec{0} + \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{Bmatrix} $$

USING SLIDE #20 FOR LINEAR ACCELERATION:

$$ {}^{1}\dot{\vec{v}}_1 = {}^{1}_{0}R\left({}^{0}\dot{\vec{\omega}}_0 \times {}^{0}\vec{P}_1 + {}^{0}\vec{\omega}_0 \times ({}^{0}\vec{\omega}_0 \times {}^{0}\vec{P}_1) + {}^{0}\dot{\vec{v}}_0\right) $$

$$ = \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\left(\vec{0} + \vec{0} + \begin{Bmatrix} 0 \\ g \\ 0 \end{Bmatrix}\right) = \begin{Bmatrix} s_1 g \\ c_1 g \\ 0 \end{Bmatrix} $$

FOR LINEAR ACCL. OF CENTER OF MASS:

$$ {}^{1}\dot{\vec{v}}_{G_1} = {}^{1}\dot{\vec{\omega}}_1 \times {}^{1}\vec{P}_{G_1} + {}^{1}\vec{\omega}_1 \times ({}^{1}\vec{\omega}_1 \times {}^{1}\vec{P}_{G_1}) + {}^{1}\dot{\vec{v}}_1 $$

$$ = \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{Bmatrix} \times \begin{Bmatrix} l_1/2 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} \times \left(\begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} \times \begin{Bmatrix} l_1/2 \\ 0 \\ 0 \end{Bmatrix} \right) + \begin{Bmatrix} s_1 g \\ c_1 g \\ 0 \end{Bmatrix} $$

$$ = \begin{Bmatrix} 0 \\ \ddot{\theta}_1 \frac{l_1}{2} \\ 0 \end{Bmatrix} + \begin{Bmatrix} -\dot{\theta}_1^2 (l_1/2) \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} s_1 g \\ c_1 g \\ 0 \end{Bmatrix} = \begin{Bmatrix} -\dot{\theta}_1^2 \left(\frac{l_1}{2}\right) + s_1 g \\ \ddot{\theta}_1 (l_1/2) + c_1 g \\ 0 \end{Bmatrix} $$

FORCE ACTING ON LINK 1

$$ {}^{1}\vec{F}_1 = m_1\,{}^{1}\dot{\vec{v}}_{G_1} = \begin{Bmatrix} -\dot{\theta}_1^2 m_1 (l_1/2) + s_1 m_1 g \\ \ddot{\theta}_1 m_1 (l_1/2) + c_1 m_1 g \\ 0 \end{Bmatrix} $$

TORQUE ACTING ON LINK 1

$$ {}^{1}\vec{N}_1 = {}^{G_1}I_1\,{}^{1}\dot{\vec{\omega}}_1 + {}^{1}\vec{\omega}_1 \times {}^{G_1}I_1\,{}^{1}\vec{\omega}_1 $$

$$ = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy_1} & 0 \\ 0 & 0 & I_{zz_1} \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} \times \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy_1} & 0 \\ 0 & 0 & I_{zz_1} \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ I_{zz_1} \ddot{\theta}_1 \end{Bmatrix} $$

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$i=1$

$$ {}^{2}\vec{\omega}_2 = {}^{2}_{1}R\,{}^{1}\vec{\omega}_1 + \dot{\theta}_2\,{}^{2}\hat{z}_2 = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{Bmatrix} $$

$$ {}^{2}\dot{\vec{\omega}}_2 = {}^{2}_{1}R\,{}^{1}\dot{\vec{\omega}}_1 + \ddot{\theta}_2\,{}^{2}\hat{z}_2 = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 + \ddot{\theta}_2 \end{Bmatrix} $$

$$ {}^{2}\dot{\vec{v}}_2 = {}^{2}_{1}R\left({}^{1}\dot{\vec{\omega}}_1 \times {}^{1}\vec{P}_2 + {}^{1}\vec{\omega}_1 \times ({}^{1}\vec{\omega}_1 \times {}^{1}\vec{P}_2) + {}^{1}\dot{\vec{v}}_1 \right) $$

$$ = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\left(\begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{Bmatrix} \times \begin{Bmatrix} l_1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} \times \left(\begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} \times \begin{Bmatrix} l_1 \\ 0 \\ 0 \end{Bmatrix} \right) + \begin{Bmatrix} s_1 g \\ c_1 g \\ 0 \end{Bmatrix} \right) $$

$$ = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{Bmatrix} -\dot{\theta}_1^2 l_1 + s_1 g \\ \ddot{\theta}_1 l_1 + c_1 g \\ 0 \end{Bmatrix} = \begin{Bmatrix} -c_2 \dot{\theta}_1^2 l_1 + s_2 \ddot{\theta}_1 l_1 + s_{12} g \\ s_2 \dot{\theta}_1^2 l_1 + c_2 \ddot{\theta}_1 l_1 + c_{12} g \\ 0 \end{Bmatrix} $$

$$ {}^{2}\dot{\vec{v}}_{G_2} = {}^{2}\dot{\vec{\omega}}_2 \times {}^{2}\vec{P}_{G_2} + {}^{2}\vec{\omega}_2 \times ({}^{2}\vec{\omega}_2 \times {}^{2}\vec{P}_{G_2}) + {}^{2}\dot{\vec{v}}_2 = $$

$$ = \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 + \ddot{\theta}_2 \end{Bmatrix} \times \begin{Bmatrix} l_2/2 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} \times \left(\begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{Bmatrix} \times \begin{Bmatrix} l_2/2 \\ 0 \\ 0 \end{Bmatrix} \right) + \begin{Bmatrix} -c_2 \dot{\theta}_1^2 l_1 + s_2 \ddot{\theta}_1 l_1 + s_{12} g \\ s_2 \dot{\theta}_1^2 l_1 + c_2 \ddot{\theta}_1 l_1 + c_{12} g \\ 0 \end{Bmatrix} $$

$$ = \begin{Bmatrix} s_2 \ddot{\theta}_1 l_1 - c_2 \dot{\theta}_1^2 l_1 - (\dot{\theta}_1 + \dot{\theta}_2)^2 l_2/2 + s_{12} g \\ c_2 \ddot{\theta}_1 l_1 + (\ddot{\theta}_1 + \ddot{\theta}_2)^2 l_2/2 + s_2 \dot{\theta}_1^2 l_2 + c_{12} g \\ 0 \end{Bmatrix} $$

$$ {}^{2}\vec{F}_2 = m_2\,{}^{2}\dot{\vec{v}}_{G_2} = m_2 \begin{Bmatrix} s_2 \ddot{\theta}_1 l_1 - c_2 \dot{\theta}_1^2 l_1 - (\dot{\theta}_1 + \dot{\theta}_2)^2 l_2/2 + s_{12} g \\ c_2 \ddot{\theta}_1 l_1 + (\ddot{\theta}_1 + \ddot{\theta}_2)^2 l_2/2 + s_2 \dot{\theta}_1^2 l_2 + c_{12} g \\ 0 \end{Bmatrix} $$

$$ {}^{2}\vec{N}_2 = {}^{G_2}I_2\,{}^{2}\dot{\vec{\omega}}_2 + {}^{2}\vec{\omega}_2 \times {}^{G_2}I_2\,{}^{2}\vec{\omega}_2 $$

$$ = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy_2} & 0 \\ 0 & 0 & I_{zz_2} \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 + \ddot{\theta}_2 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{Bmatrix} \times \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{yy_2} & 0 \\ 0 & 0 & I_{zz_2} \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ (\ddot{\theta}_1 + \ddot{\theta}_2)\, I_{zz_2} \end{Bmatrix} $$

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INWARD FORCE ITERATIONS: $i = 2 \rightarrow 1$

FROM SLIDE 22:

$i=2$

$$ {}^{2}\vec{f}_2 = {}^{2}_{3}R\, \cancel{{}^{3}\vec{f}_3}^{\,0} + {}^{2}\vec{F}_2 = m_2 \begin{Bmatrix} s_2\ddot{\theta}_1 l_1 - c_2\dot{\theta}_1^2 l_1 - (\dot{\theta}_1+\dot{\theta}_2)^2 l_2/2 + s_{12} g \\ c_2\ddot{\theta}_1 l_1 + (\ddot{\theta}_1+\ddot{\theta}_2)^2 l_2/2 + s_2\dot{\theta}_1^2 l_2 + c_{12} g \\ 0 \end{Bmatrix} $$

$$ {}^{2}\vec{n}_2 = {}^{2}\vec{N}_2 + {}^{2}_{3}R\, \cancel{{}^{3}\vec{n}_3}^{\,0} + {}^{2}\vec{P}_{G_2} \times {}^{2}\vec{F}_2 + {}^{2}\vec{P}_3 \times {}^{2}_{3}R\, \cancel{{}^{3}\vec{f}_3}^{\,0} $$

$$ = \begin{Bmatrix} 0 \\ 0 \\ (\ddot{\theta}_1+\ddot{\theta}_2) I_{zz_2} \end{Bmatrix} + \begin{Bmatrix} l_2/2 \\ 0 \\ 0 \end{Bmatrix} \times {}^{2}\vec{F}_2 $$

FOR REVOLUTE JOINT

$$ = \begin{Bmatrix} 0 \\ 0 \\ (\ddot{\theta}_1+\ddot{\theta}_2) I_{zz_2} + (c_2 l_1 + l_2/2)\frac{l_2}{2} m_2 \ddot{\theta}_1 + \frac{l_2^2}{4} m_2 \ddot{\theta}_2 + s_2 l_1 \frac{l_2}{2} m_2 \dot{\theta}_1^2 + \frac{l_2}{2} c_{12} m_2 g \end{Bmatrix} $$

$$ \tau_2 = {}^{2}\vec{n}_2^{\,T}\, {}^{2}\hat{z}_2 = (\ddot{\theta}_1+\ddot{\theta}_2) I_{zz_2} + (c_2 l_1 + l_2/2)\frac{l_2}{2} m_2 \ddot{\theta}_1 + \frac{l_2^2}{4} m_2 \ddot{\theta}_2 + s_2 l_1 \frac{l_2}{2} m_2 \dot{\theta}_1^2 + \frac{l_2}{2} c_{12} m_2 g $$

EQ 1

$i=1$

$$ {}^{1}\vec{f}_1 = {}^{1}_{2}R\, {}^{2}\vec{f}_2 + {}^{1}\vec{F}_1 = \begin{bmatrix} c_2 & -s_2 & 0 \\ s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^{2}\vec{f}_2 + \begin{Bmatrix} -(l_1/2) m_1 \dot{\theta}_1^2 + s_1 m_1 g \\ (l_1/2) m_1 \ddot{\theta}_1 + c_1 m_1 g \\ 0 \end{Bmatrix} $$

$$ = \begin{Bmatrix} -l_1 m_2 \dot{\theta}_1^2 - c_2 \frac{l_2}{2} m_2 (\dot{\theta}_1+\dot{\theta}_2)^2 + s_2 \frac{l_2}{2} m_2 (\ddot{\theta}_1+\ddot{\theta}_2) - \frac{l_1}{2} m_1 \dot{\theta}_1^2 + s_1 m_1 g + s_1 m_2 g \\ l_1 m_2 \ddot{\theta}_1 - s_2 \frac{l_2}{2} m_2 (\dot{\theta}_1+\dot{\theta}_2)^2 + c_2 \frac{l_2}{2} m_2 (\ddot{\theta}_1+\ddot{\theta}_2) + \frac{l_1}{2} m_1 \ddot{\theta}_1 + c_1 m_1 g + c_1 m_2 g \\ 0 \end{Bmatrix} $$

$$ {}^{1}\vec{n}_1 = {}^{1}\vec{N}_1 + {}^{1}_{2}R\, {}^{1}\vec{n}_2 + {}^{1}\vec{P}_{G_1} \times {}^{1}\vec{F}_1 + {}^{1}\vec{P}_2 \times {}^{1}_{2}R\, {}^{2}\vec{f}_2 $$

$$ = \begin{Bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 I_{zz_1} \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ \tau_2 \end{Bmatrix} + \begin{Bmatrix} l_1/2 \\ 0 \\ 0 \end{Bmatrix} \times {}^{1}\vec{F}_1 + \begin{Bmatrix} l_1 \\ 0 \\ 0 \end{Bmatrix} \times {}^{1}_{2}R\, {}^{2}\vec{f}_2 $$

$$ = \begin{Bmatrix} 0 \\ 0 \\ I_{zz_1}\ddot{\theta}_1 + \tau_2 + \ddot{\theta}_1 m_1 \frac{l_1^2}{4} + c_1 \frac{l_1}{2} m_1 g + c_1 l_1 m_2 g + l_1^2 m_2 \ddot{\theta}_1 - \frac{s_2 l_2 l_1}{2} m_2 (\dot{\theta}_1+\dot{\theta}_2)^2 + c_2 \frac{l_2 l_1}{2} m_2 (\ddot{\theta}_1+\ddot{\theta}_2) \end{Bmatrix} $$

$$ \tau_1 = {}^{1}\vec{n}_1^{\,T}\, {}^{1}\hat{z}_1 = I_{zz_1}\ddot{\theta}_1 + \tau_2 + \ddot{\theta}_1 m_1 \frac{l_1^2}{4} + c_1 \frac{l_1}{2} m_1 g + c_2 l_1 m_2 g + l_1^2 m_2 \ddot{\theta}_1 - s_2 \frac{l_2 l_1}{2} m_2 (\dot{\theta}_1+\dot{\theta}_2)^2 + c_2 \frac{l_2 l_1}{2} m_2 (\ddot{\theta}_1+\ddot{\theta}_2) $$

EQ 2

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WRITING THE JOINT TORQUES (FROM EQ'S ① & ②) IN MATRIX FORM:

$$\begin{Bmatrix} \tau_1 \\ \tau_2 \end{Bmatrix} = \overbrace{\begin{bmatrix} I_{zz_1} + I_{zz_2} + \frac{1}{4} l_2^2 m_2 + l_1 l_2 c_2 m_2 + \frac{1}{4} l_1^2 m_1 + m_2 & I_{zz_2} + \frac{1}{4} m_2 l_2^2 + \frac{1}{2} m_2 c_2 l_1 l_2 \\ I_{zz_2} + \frac{1}{4} m_2 l_2^2 + \frac{1}{2} l_1 l_2 c_2 m_2 & I_{zz_2} + \frac{l_2^2}{4} m_2 \end{bmatrix}}^{M(\vec{\theta})} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix}$$

$$+ \underbrace{\begin{Bmatrix} -\frac{1}{2} l_1 l_2 s_2 m_2 \dot{\theta}_2^2 - l_1 l_2 s_2 m_2 \dot{\theta}_1 \dot{\theta}_2 \\ \frac{1}{2} l_1 l_2 s_2 m_2 \dot{\theta}_1^2 \end{Bmatrix}}_{\vec{V}(\vec{\theta}\ \dot{\vec{\theta}})} + \underbrace{\begin{Bmatrix} \left(\frac{1}{2} l_2 c_{12} m_2 + \frac{1}{2} l_1 c_1 m_1 + l_1 c_1 m_2\right) g \\ \frac{1}{2} l_2 c_{12} m_2 g \end{Bmatrix}}_{\vec{G}(\vec{\theta})}$$

THE DYNAMIC EQUATIONS BECOME:

$$\vec{\tau} = [M(\vec{\theta})]\, \ddot{\vec{\theta}} + \vec{V}(\vec{\theta}, \dot{\vec{\theta}}) + \vec{G}(\vec{\theta})$$

WHERE

$[M(\vec{\theta})]$ IS THE $\underline{n \times n}$ MASS MATRIX

$\vec{V}(\vec{\theta}, \dot{\vec{\theta}})$ IS AN $\underline{n \times 1}$ VECTOR, WHICH INCLUDES CENTRIFUGAL AND CORIOLIS TERMS

$\vec{G}(\vec{\theta})$ IS AN $\underline{n \times 1}$ VECTOR, WHICH INCLUDES THE GRAVITATIONAL TERMS.

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